

Reflective Journal

Group Members Names:

Edwin Alvarado, Kala Hayes, Anashrah Adil, Zain Ahmed

Class: 6263-ITAI-1371 Intro to Machine Learning-S10-12321

In this lab, we focused on moving beyond raw accuracy to evaluate machine learning models using more reliable metrics, including the confusion matrix, precision, recall, and cross-validation.

Working through the corrupted notebook file was an immediate real-world debugging lesson. Resolving execution errors and restoring broken cell structures taught us the importance of understanding notebook underlying structures and inspecting environment execution order rather than assuming a file works out-of-the-box.

Once the code ran, examining the confusion matrix revealed critical insights that the surface-level **80.00%** accuracy hid. While high accuracy made the model look solid at a glance, the matrix showed us the precise breakdown of **20 false negatives** and **15 false positives**. This proved that high accuracy can mask significant misclassifications—a lesson directly connected to our stroke prediction work, where missing a high-risk patient (a false negative) carries life-threatening consequences compared to simply misclassifying a healthy patient.

Finally, comparing our single train-test split accuracy (**80.00%**) against our 5-fold cross-validation average (**78.68%**, ranging from **77.53% to 80.34%**) fundamentally changed how we will report performance moving forward. A single split can easily suffer from a "lucky" random split. Going forward, we will report cross-validation means and standard deviations to provide a realistic, honest metric of generalized model performance rather than relying on an overly optimistic single run.